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//SIT 55 : ASSIGNMENT 7 : TEMPLATE
/*Implement a generic program using any collection class to count the number of elements in a
collection that have a specific property such as even numbers, odd number, prime number and
palindromes.*/

import java.util.*;
public class GenericProg {
    static int count = 0; //COUNT VARIABLE

    //FUNCTION TO CHECK PALINDROME
    static void check_palindrome(String x){
        StringBuilder s1 = new StringBuilder(x);
        if(x.equals(s1.reverse().toString())){
            System.out.println(x+" is a Palindrome");
            count += 1; //count the number of palindromes
        }
        else{
            System.out.println(x+" is not a Palindrome");
        }
    }
        //FUNCTION TO CHECK EVEN OR ODD
    static void even_odd(int x){
        if(x % 2 == 0){
            System.out.println(x+" IS EVEN");
            count += 1; //count the number of even numbers
        }
        else{
            System.out.println(x+" IS ODD");
        }
    }
//FUNCION TO CHECK PRIME NUMBER
    static void prime(int x){
        boolean flag = false;
        for(int i = 2; i <= x/2; ++i){
            if(x % i == 0){
                flag = true;
                break;
            }
        }
        if (!flag){
            System.out.println(x + " is a prime number.");
            count += 1; //count the number of prime numbers
        }
        else{
            System.out.println(x + " is not a prime number.");
        }
    }
    //FUNCTION TO DECIDE WHICH FUNCTION TO CHECK
    static void check(int ch,int x){
        switch(ch){
            case 1:
                even_odd(x); //call even_odd fucntion for number x
                break;
            case 2:
                prime(x); //call prime fucntion for number x
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            break;
         default:
            System.out.println("ENTER CORRECT OPTION");
      }
   } //FUNCTION FOR INTEGER ARRAY
   static void number_op(){
      int element,n,choice;

      Scanner sc = new Scanner(System.in);

      //ArrayList from Collection Interface
      //Integer type
      ArrayList<Integer> nums = new ArrayList<Integer>();

      System.out.println("Enter the number of elements:");
      n = sc.nextInt();
      System.out.println("Enter the elements:");

      for(int i=0;i<n;i++){
         element = sc.nextInt();
         nums.add(element); //Add elements to the ArrayList
      }

      System.out.println("Enter the Operation to be performed:");
      System.out.println("1. ODD or EVEN");
      System.out.println("2. PRIME OR NOT");
      choice = sc.nextInt();

      Iterator itr = nums.iterator(); //Iterator from the COLLECTION interface
      count = 0;
      while(itr.hasNext()){ //Loop till there are elements in the ArrayList
        check(choice,(int)itr.next()); //call the check function for each element
      }

      //Give the Count
      if(choice == 1){
         System.out.println("The number of EVEN numbers is: "+ count);
         System.out.println("The number of ODD numbers is: "+ (nums.size()-count));
      }
      else{
         System.out.println("The number of PRIME numbers is: "+ count);
         System.out.println("The number of Non-PRIME numbers is: "+ (nums.size()-count));
      }
   }

//FUNCTION FOR STRING ARRAY
   static void string_op(){
      int n;
      String word;

      //ArrayList from COLLECTION interface
      //String type
      ArrayList<String> words = new ArrayList<String>();

      Scanner sc = new Scanner(System.in);
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        System.out.println("Enter the number of elements:");
        n = sc.nextInt();
        System.out.println("Enter elements:");
        for(int i=0;i<n;i++){
            word = sc.next();
            words.add(word); //Add elements to the ArrayList
        }

        count = 0;
        for(String w:words){ //Loop the ArrayList
            check_palindrome(w);
        }

        System.out.println("The number of PALINDROMES is: "+ count);

    }
 public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        //Choose the type of List needed
        System.out.println("Choose Type:");
        System.out.println("1. String");
        System.out.println("2. Integer");
        int ch = sc.nextInt();

        if(ch == 2)
            number_op(); //Calls Interger arraylist
        else
            string_op(); //Calls String arraylist
    }

}


/*OUTPUT:
CASE 1]
Choose Type:
1. String
2. Integer
1
Enter the number of elements:
3
Enter elements:
66
21
33
66 is a Palindrome
21 is not a Palindrome
33 is a Palindrome
The number of PALINDROMES is: 2

CASE 2]
Choose Type:
1. String
2. Integer
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2
Enter the number of elements:
4
Enter the elements:
22
13
33
97
Enter the Operation to be performed:
1. ODD or EVEN
2. PRIME OR NOT
2
22 is not a prime number.
13 is a prime number.
33 is not a prime number.
97 is a prime number.
The number of PRIME numbers is: 2
The number of Non-PRIME numbers is: 2*/